Farhana Najnin Mahi



AI/ML Researcher | AI Automation Engineer

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CAREER OBJECTIVE

Computer Science and Engineering graduate from Ahsanullah University of Science and Technology with hands-on experience building LLM-orchestrated automation agents (Claude Skills) for live production environments, alongside machine learning, NLP, and full-stack development. Skilled in workflow automation, REST APIs, RAG pipelines, data-pipeline validation, and Power BI dashboards. Interested in building practical, production-grade AI automation systems that create measurable operational impact.

EDUCATION

Ahsanullah University of Science and Technology
Nov 2021 - Dec 2025

B.Sc. in Computer Science and Engineering | Tejgaon, Dhaka

Dhaka City College
2018 - 2020

Higher Secondary Certificate (HSC) | GPA: 5.00 | Dhanmondi, Dhaka

Dagonbhuiyan Girls High School
2016 - 2018

Secondary School Certificate (SSC) | GPA: 5.00 | Dagonbhuiyan, Feni

EXPERIENCE

Executive — Foodi Express Limited
Aug 2026 - Present

US-Bangla Group Concern

Shops Department · Onsite · Data Analysis · Dashboard Management · Data Handling

- Support shop/vendor onboarding by verifying shop details and menu/product listings for accuracy, and identifying duplicate or inconsistent shop entries.
- Monitor shop-level performance (order volume, ratings, cancellation rate) to identify best- and worst-performing shops and flag inactive or underperforming shops for follow-up.
- Audit product pricing for consistency and accuracy, monitor stock and availability status, and compare pricing against competitors where applicable.
- Prepare daily, weekly, and monthly Excel reports, trackers, and pivot-based summaries for KPI monitoring, sales and stock visibility, and management decision-making.
- Identify data inconsistencies and frontend-facing errors in pricing, availability, and shop information, and coordinate with shop owners/vendors to correct them.
- Track shop-wise revenue and commission figures to support financial visibility and reporting.
- Analyze shop-wise performance of promotional and discount campaigns to support marketing and business decisions.

Research Assistant | ELITE Research Lab
Jul 2026 - Present

Remote · Promoted from Research Intern on Nov 12, 2026 · AI/ML Research · LLM/NLP

- Assist with AI/ML and LLM/NLP research activities including literature review, paper reading, and research note preparation.
- Support experiment planning, result interpretation, and structured documentation for ongoing research tasks.
- Contribute to technical reporting by organizing findings, summarizing papers, and maintaining reproducible research records.

AI Automation Intern | Confidence Infrastructure PLC
Jun 2026 - Aug 2026

Scalability Team (AI Automation) · Onsite · Supply Chain Cashflow & SCM Dashboards · LLM-Orchestrated Agents (Claude Skills) · Workflow Automation

- Built and piloted two Claude-Skills-based automation agents for cashflow and SCM dashboards using Python (pandas, openpyxl, reportlab, python-docx).
- **Cashflow Agent:** Developed cashflow-mismatch forecasting (4/8/12-week horizons) and project-speed scoring across a live portfolio of EPC/infrastructure tender projects; analyzed projected portfolio-wide funding gaps to inform risk assessment and planning.
- **SCM Agent:** Automated PR cut-off tracking, PR-to-PO conversion, in-transit/overdue shipment monitoring, inventory reorder alerts, delivery-status tracking, and monthly fund/shortage forecasting with automated management reports.

- Developed schedule-performance scoring and data-quality auditing for operational exports; investigated reporting inconsistencies, documented root causes, and recommended workflow improvements.
- Validated cashflow and SCM pipelines through automated verification checks and regression tests across source Excel files to assess data accuracy, forecasting consistency, and report reliability.
- Built a staff lunch-booking and spend-tracking feature using Laravel/JavaScript through AI-assisted development in Cursor Pro; directed implementation and tested application workflows.

RESEARCH PUBLICATIONS

Big Data Analysis for Forest Cover Prediction with PySpark

IEEE QPAIN 2026 - Chittagong, Bangladesh | DOI: 10.1109/QPAIN69676.2026.11546446

- Conducted large-scale forest cover prediction using PySpark distributed ML (Logistic Regression, Random Forest, SVM, MLP, CatBoost) with SMOTE class-imbalance handling; achieved 96.18% classification accuracy with CatBoost.
- Co-authors: Meherunnesa Hossain Ibnath, Sadia Sabrin Neha, Nusrat Jahan.

SNR-Aware Noise-Robust Keyword Spotting for Low-Resource Voice Command Recognition using Multi-Feature Hybrid CNN-BIGRU

IEEE SPICSCON 2026 - Conference | Published | DOI pending

- Proposed an SNR-aware, noise-robust keyword-spotting model fusing multiple acoustic features for reliable low-resource voice command recognition under noisy conditions.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Uncertainty-Calibrated Learning Analytics for Learner Risk Support: Selective Prediction and Human-in-the-Loop Triage Using OULAD

IEEE MIET 2026 - Conference | Published | DOI pending

- Developed an uncertainty-calibrated learning-analytics framework using selective prediction and human-in-the-loop triage to support early identification of at-risk learners on the OULAD dataset.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Time-Aware and Explainable Hybrid Ensemble Learning for PM10 Prediction Using Meteorological and Lag-Based Features

IEEE MIET 2026 - Conference | Published | DOI pending

- Built a time-aware, explainable hybrid ensemble for PM10 air-quality prediction using meteorological and lag-based features.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Cross-Domain Toxicity Detection with Binary Label Harmonization: A Comparative Study of Fine-Tuned Transformers, Prompt-Based Language Models, and Hybrid Ensembles

IEEE SPICSCON 2026 - Conference | Published | DOI pending

- Conducted a cross-domain toxicity-detection study with binary label harmonization, comparing fine-tuned transformers, prompt-based language models, and hybrid ensembles.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

ACHIEVEMENTS

- 6th Place - RSBRU International Research Summit 2026: presented "Explainable Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Grad-CAM" at Begum Rokeya University, Bangladesh.
- 8th Position (AUST) - AgentX: The AI Challenge 2025, NetCom Learning Bangladesh (Powered by Microsoft): competed in Bangladesh's largest AI prompting competition, covering Microsoft 365 Copilot, AI+ Prompt Engineering, and AI Agent Design/Creation; Prize-Giving Ceremony hosted at Ahsanullah University of Science and Technology.
- Campus Ambassador - ICT Olympiad Bangladesh (2022): represented AUST and promoted national-level ICT participation.

PROJECTS

FastAPI Resume Matcher | GitHub

FastAPI · SentenceTransformers · Scikit-learn · Python

- Developed a RESTful API using MiniLM sentence embeddings for semantic resume-job matching; added skill extraction, gap analysis, and Swagger UI testing.

ClinReasoner - Guideline-Grounded Clinical Decision Agent | GitHub

RAG · FAISS · Sentence-Transformers · Streamlit · Python

- Built a multi-domain clinical guideline RAG system using PyMuPDF and FAISS with rule-based risk detection and citation-backed recommendations.

ChurnBuster - Customer Churn Prediction & Retention Dashboard | GitHub

XGBoost · SHAP · Streamlit · Python

- Developed an end-to-end churn prediction dashboard with probability-based risk scores, cost-based threshold optimisation, and SHAP explainability.

Mobile Financial Service Campaign Performance Dashboard | GitHub

Power BI · Power Query · DAX · CSV Dataset · Data Visualization · KPI Reporting

- Designed an interactive Power BI dashboard analyzing revenue, cost, ROI, conversion rate, transactions, regional performance, and monthly trends using DAX KPIs and slicers.

Skin Disease Detection System | GitHub

CNN · FastAPI · Streamlit · Python · Kaggle Skin Disease Dataset

- Built a CNN-based multi-class skin disease classifier for 10 disease categories using 27k+ images; developed a FastAPI backend and Streamlit UI with image preprocessing and rule-based recommendations.

Uncertainty-Aware Retail Demand Forecaster | Retail Repo | Uncertainty Repo

Prophet · Streamlit · Scikit-learn · Python

- Built store-level demand forecasting with P50/P90 uncertainty estimates and inventory simulation; compared ordering strategies using MAE and cost-based metrics.

ECourse - Full-Stack E-Learning Platform | GitHub

Django · PostgreSQL · Bootstrap · Python

- Designed a Django e-learning platform with user auth, course creation, enrollment, progress tracking, role-based access, PostgreSQL schema, and responsive Bootstrap UI.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, JavaScript, PHP

Automation & AI Agents: Claude Skills / Agentic Workflows, LLM Orchestration, Cursor (AI pair-programming), Prompt Engineering, Workflow Automation, Process Automation, Data Pipeline Validation, Excel/Report Automation (openpyxl, python-docx, reportlab)

AI / ML: Machine Learning, Deep Learning, CNN, XGBoost, Scikit-learn, BanglaBERT, XLM-RoBERTa, mBERT, Prophet, SHAP, FAISS, LoRA/PEFT, PySpark, SMOTE

Research & NLP: LLM/NLP, Literature Review, Paper Analysis, RAG, Transformer Models, Experiment Documentation

Backend & Frameworks: FastAPI, Django (MTV/ORM), Streamlit, REST API Design

Data & Analytics: Pandas, NumPy, Matplotlib, Excel, Power BI, Power Query, DAX, KPI Reporting

Databases: PostgreSQL, MySQL

Tools & Platforms: Git & GitHub, Google Colab, VS Code, FastAPI/Uvicorn, ngrok, PyMuPDF, Render, Android Studio

CERTIFICATIONS & TRAINING

- AI+ Prompt Engineer Level 1™ - AI CERTs™
- Microsoft Azure AI Fundamentals (AI-900) - Microsoft
- Introduction to Generative AI & Image Generation - Google Cloud
- Machine Learning Specialization - Andrew Ng, Coursera
- Google Certified Educator: Level 1 - Google for Education
- Student Ambassador Technical Training Program - Microsoft
- Hands-on Workshop: FPGA Acceleration for ML Applications - AUST (Jan 2025)
- Hands-on Workshop: AWS, CNCF & GitHub - AUSTPIC (Aug 2025)

REFERENCES

Mohammad Mosharaf Hossain
DGM (Zonal Manager)
Grameen Bank
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